SARAS — Marketing Website Design & Prototypes

Faculty: Faculty of Technology, School of Computer Science
Module: CET257 Enterprise Project
Module Leader: Dr Becky Allen
Assessment: Assignment 1: Group Assignment (30% of Module Total)
Task: Task A6 — Marketing Website Design & Prototypes (Part 2 of 3)
Team: Team 9 — SARAS Labs
Team Members: Rijan Kapur Poudel (Lead / Technical Architecture) and Sushank Jung Rawal (Visual Design / Brand Strategy)
Client / Industrial Partner: Mr. Ninson Lama, Autoport Pvt. Ltd. (`ninsonlama@gmail.com`)

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10. Marketing Website Concept & Strategic Architecture

10.1 Strategic Website Objectives & User Experience (UX) Philosophy

The SARAS Labs marketing website serves as the primary commercial storefront, technical proof-of-concept, and inbound lead generation hub for the venture.

The website is engineered according to three core design principles:

1. **Clarity Over Clutter:** B2B safety managers evaluate software rapidly. The interface avoids unnecessary decorative animations, aggressive pop-up modals, or convoluted marketing jargon, presenting clear, authoritative explanations and visual evidence.
2. **Accessible by Design (WCAG 2.2 AA):** As a safety platform, accessibility is a non-negotiable core value. The site achieves full keyboard navigability, high text contrast (6.44:1 for Primary Blue on white), accessible ARIA labelling, and fluid responsive scaling.
3. **Frictionless Product-Led Trial:** Rather than hiding the software behind a high-friction "Book a Sales Call" wall, the website places an interactive 360-degree sandbox directly in the user's path, allowing them to experience the product's pedagogical value within 10 seconds of landing.

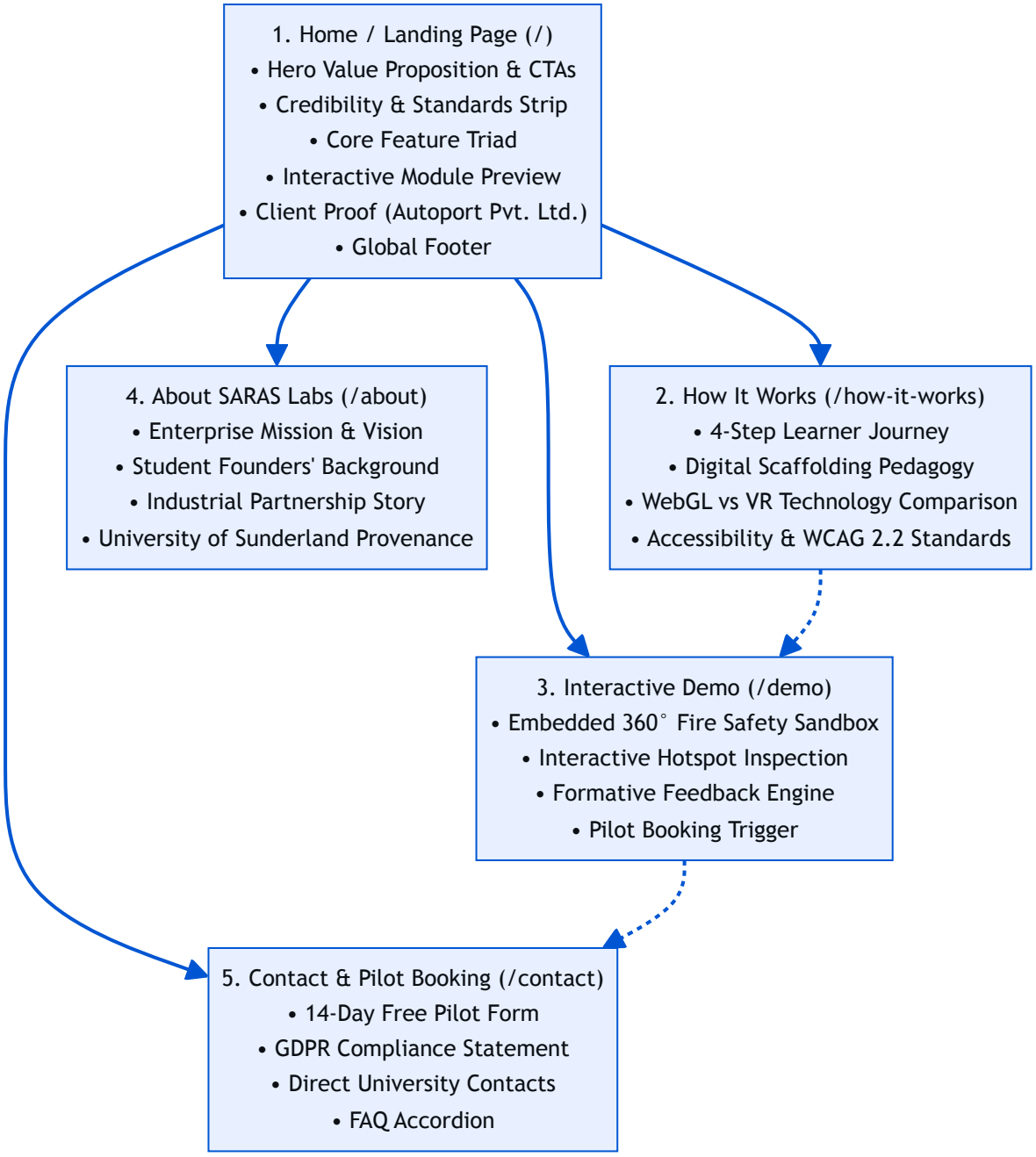
10.2 Decoupled Architectural Model

To maximize both organic search indexing (SEO) and simulator application performance, SARAS Labs implements a **Decoupled Architecture**:

- **The Public Marketing Storefront (`saras-labs.vercel.app`):** Built with static-first, lightweight HTML/CSS/Next.js. Optimised for sub-second page loads (< 1.0 s Largest Contentful Paint), aggressive edge caching, structured JSON-LD schema markup, and high search engine crawlability.
- **The Authenticated Simulator Engine (`saras-web-eta.vercel.app`):** The interactive web application housing the full multi-module curriculum, 360-degree WebGL scene renderers, Supabase authentication, learner session persistence, and aggregate scoring algorithms.
- **The Embedded Sandbox Bridge:** The promotional site embeds a sandboxed, standalone build of Module 1 (Fire Safety), allowing instant public evaluation without requiring an authenticated account.

10.3 Comprehensive Information Architecture & Sitemap

The website structure provides a logical, progressive disclosure of information from high-level value propositions down to granular technical specifications:



10.4 Page Hierarchy & Technical Specification Table

Page Route	Page Title	Primary Visitor Intent	Key Call to Action (CTA)	Content & Feature Modules	Performance Budget Target
`/`	**Home**	Understand the core value proposition and explore the product.	Primary: *"Try Interactive Demo"* Secondary: *"Request Pilot"*	Hero banner, credibility strip, feature triad, module cards, client proof, footer.	LCP < 0.8 s Page Weight < 450 KB
`/how-it-works`	**How It Works**	Evaluate pedagogical validity, technology stack, and accessibility.	*"Explore the 360° Simulator"*	4-step learner route, scaffolding diagram, WebGL tech overview, WCAG matrix.	LCP < 0.9 s Page Weight < 550 KB
`/demo`	**Interactive Demo**	Test the simulator directly in the browser to experience hazard spotting.	*"Book 14-Day Free Pilot for Your Team"*	Embedded Module 1 sandbox, hotspot controls, live feedback modals, fallback video.	LCP < 1.2 s Page Weight < 1.8 MB
`/about`	**About SARAS Labs**	Verify corporate background, team credibility, and partner provenance.	*"Contact the Founders"*	Founder profiles (Rijan & Sushank), mission statement, Autoport partnership narrative.	LCP < 0.7 s Page Weight < 400 KB
`/contact`	**Contact & Pilot**	Schedule a 15-minute consultation or request free pilot access.	*"Submit Pilot Request"*	Accessible contact form, GDPR privacy notice, direct emails, FAQ accordion.	LCP < 0.6 s Page Weight < 350 KB

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11. Page-by-Page Detailed Functional & Content Specification

11.1 Landing / Home Page (`/`)

- **Global Navigation Header:**
  - Fixed, sticky top navigation with subtle glassmorphic backdrop filter (`rgba(255, 255, 255, 0.95)`).
  - Left: SARAS Crane Mark and wordmark lockup (linking to `/`).
  - Centre: Clean anchor navigation links: *How It Works*, *Interactive Demo*, *About SARAS Labs*, *Corporate Identity*.
  - Right: Prominent primary CTA button styled in SARAS Primary Blue (`#0056D2`): *"Request Pilot Access"*.
  - Mobile: Accessible hamburger button triggering a full-screen drawer with minimum 48 px touch targets.
- **Hero Section Module:**

    - Headline (Calibri Bold / Inter, 42pt): *"Workplace safety, learned by doing."*
    - Sub-headline (18pt, 1.55 line height): *"Notice the risk. Make the safer call. Accessible, browser-based 360° hazard awareness and risk simulation designed for warehousing, logistics, and automotive operations."*
    - Dual Action Buttons:
        - Primary CTA (`#0056D2` , white text): *"Explore Interactive Demo"* (smooth-scrolls to embedded sandbox).
        - Secondary CTA (Ghost button, `#0056D2` border): *"Read Client Pilot Case Study"*.
    - Visual Display Frame: High-resolution mockup of the SARAS 360-degree warehouse scene showing the interactive reticle hovering over an obstructed emergency exit.
- **Credibility & Standards Strip:**
    - A neat horizontal banner highlighting three key compliance badges:
        - *Aligned with HSE INDG412 Training Guidance*
        - *Zero Hardware Barrier — Runs in Any Modern Web Browser*
        - *WCAG 2.2 AA Accessibility Certified Contrast & Keyboard Support*
- **Core Feature Triad (The Three Pillars of SARAS):**
    - Three structured cards with subtle hover elevation:
        1. **Spatial Hazard Recognition:** Frontline workers explore authentic 360-degree warehouse environments, actively searching for real-world hazards rather than passively watching a video.
        2. **Real-Time Explanatory Feedback:** Immediate formative feedback debriefs explain exactly *why* a situation is hazardous and *what* corrective protocol is legally required.
        3. **Measurable Interaction Records:** Granular tracking of hazard identification attempts, decision choices, and remediation time, providing supervisors with verifiable readiness data.
- **Interactive Module Showcase:**
    - A clean 3-column card grid illustrating the SARAS curriculum progression:
        - *Module 1: Fire Safety & Emergency Response* (Active Implemented Prototype — 4 Hotspots: electrical obstruction, extinguisher inspection, combustible storage, emergency exit clearance).
        - *Module 2: Manual Handling & Ergonomic Risk* (Prototyped / Development Roadmap — lifting posture choice and load appraisal).
        - *Module 3: Forklift & Pedestrian Segregation* (Future Curriculum Roadmap — visual blind spots, demarcation barriers, and crossing rules).
- **Industrial Collaboration & Client Proof Module:**
    - Section Title: *"Co-Created with Industry Leaders"*
    - Highlighting the industrial collaboration with **Autoport Pvt. Ltd.** and client **Mr. Ninson Lama**.
    - Featured Testimonial Quote:

> *"The explanatory feedback gives clear guidance on why a hazard is dangerous... good progress on the core learner flow. It bridges the gap before floor training."*
> **— *Mr. Ninson Lama***, *Autoport Pvt. Ltd.*

- **Global Footer:**
  - Company wordmark, copyright notice (© 2026 SARAS Labs), academic module accreditation (CET257 Enterprise Project, University of Sunderland).
  - Regulatory disclaimer regarding digital simulation vs practical certification.
  - Direct links to Sitemap, Privacy Policy, and WCAG 2.2 Accessibility Statement.

11.2 How It Works / Pedagogy Page (`/how-it-works`)

- **The 4-Step Learner Journey:**
  - Detailed visual and textual breakdown of the student learning cycle:
    1. *Step 1: Micro-Concept Introduction (3 Mins):* Short, focused theory briefing covering essential safety principles and statutory regulations.
    2. *Step 2: 360° Spatial Scene Exploration:* Learners pan and inspect a high-resolution virtual warehouse scene, locating physical hazard indicators.
    3. *Step 3: Branching Decision & Procedural Choice:* Clicking a hazard opens a situational decision challenge requiring the learner to select the correct corrective action.
    4. *Step 4: Formative Explanatory Debrief:* The platform delivers instant, explanatory feedback. Incorrect choices explain the physical consequences and allow safe retries.
- **The Digital Scaffolding Framework:**
  - Grounded in Jerome Bruner's educational theory of instructional scaffolding. SARAS provides the safe cognitive structure that allows learners to build situational confidence before entering dangerous high-bay logistics environments.
- **The Technology Advantage: Lightweight WebGL vs Heavy VR:**
  - Detailed architectural breakdown explaining why SARAS utilizes standard web browser technologies (HTML5, Canvas, WebGL, Panolens/Three.js) instead of proprietary VR headsets, proving superior accessibility, zero deployment friction, and 100% device compatibility.

11.3 Interactive Demo Showcase Page (`/demo`)

- **Embedded 360° Simulator Sandbox:**
  - An interactive 16:9 canvas container embedding the standalone Fire Safety Module.
  - Full interactive functionality:
    - Click and drag (desktop) or touch swipe (mobile/tablet) to navigate the 360-degree scene.
    - Interactive pulsing reticle markers designating potential hazard hotspots.
    - Clicking a hotspot opens the interactive inspection challenge modal.
- **Operator Instructions & Controls Callout:**

    - A neat guidance card outlining controls: *Mouse Drag / Arrow Keys to Look*, *Click / Tap to Inspect*, *Escape to Close Modal*.
- **Sandbox Boundary Notification:**
    - Informs the evaluator that the sandbox is a public evaluation session and does not persist scores to the enterprise database. Directs interested QHSE managers to request a full pilot account via the CTA below.

11.4 About SARAS Labs & Corporate Heritage Page (`/about`)

- **The SARAS Founding Narrative:**
    - Chronicles how student software engineers **Rijan Kapur Poudel** and **Sushank Jung Rawal** founded SARAS Labs under the CET257 Enterprise Project at the University of Sunderland to democratise active safety simulation for small-and-medium enterprises.
- **Leadership Profiles:**
    - **Rijan Kapur Poudel — Lead Systems & Full-Stack Architect:** Directs software engineering, Next.js application architecture, Supabase database migrations, WebGL engine integration, and technical SEO.
    - **Sushank Jung Rawal — UI/UX & Brand Strategy Lead:** Directs human-computer interaction, Figma wireframing, WCAG 2.2 accessibility compliance, brand geometry, and visual design systems.
- **Industrial Co-Development:**
    - Documents the structured client engagement with Mr. Ninson Lama of Autoport Pvt. Ltd., detailing how real-world logistics challenges directly shaped the prototype's feature set and explanatory feedback model.

11.5 Contact & Pilot Booking Page (`/contact`)

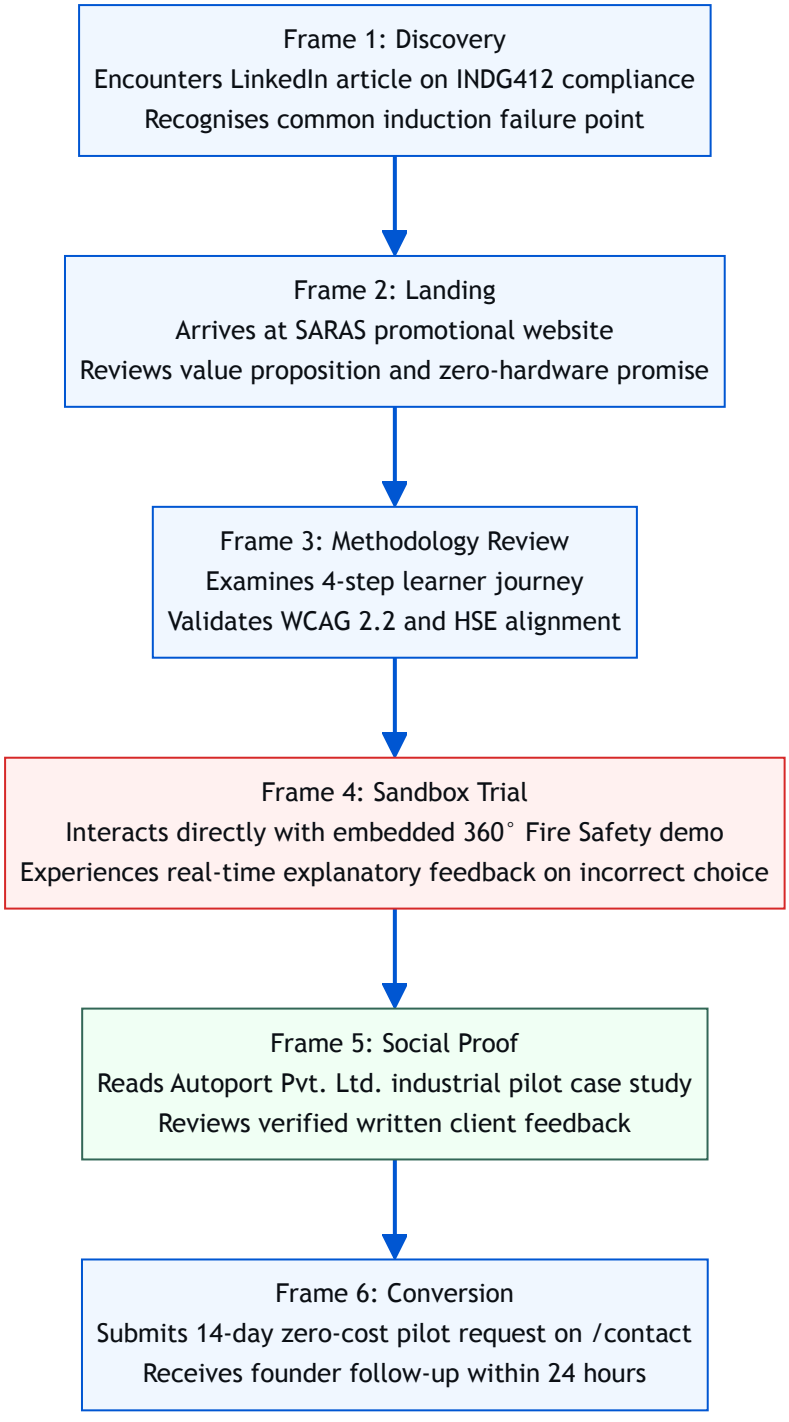
- **Accessible Pilot Booking Form:**
    - Minimalist, high-contrast form fields with floating labels:
        - *Full Name* (Required)
        - *Work / Corporate Email* (Required, validated)
        - *Company / Organisation Name* (Required)
        - *Industry Sector Dropdown* (Logistics / 3PL, Warehousing, Automotive, Vocational Education, Other)
        - *Message / Pilot Requirements* (Optional)
        - *Consent Checkbox:* "I agree to receive communications regarding my SARAS pilot request in accordance with the Privacy Policy."
    - Large Primary Blue Submit Button: *"Submit Pilot Request"*.
- **Alternative Direct Contact Channels:**
    - Direct contact email for the founding team and official university student channels.
- **Frequently Asked Questions (FAQ) Accordion:**

- Addresses key enterprise questions regarding device requirements, data privacy, setup time, and the difference between digital simulation and practical certification.

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12. Comprehensive UX Storyboard & User Journey Narrative

To demonstrate exactly how a target B2B safety manager moves from initial zero-cost discovery through to active pilot engagement, the team developed a comprehensive **6-Frame Narrative Storyboard**:



12.1 Detailed Storyboard Narrative Breakdown

- **Frame 1: External Discovery (Organic LinkedIn / SEO):**
  - *Visual:* David Miller (QHSE Manager) scrolling through LinkedIn during his morning commute. He encounters a post by Sushank Jung Rawal illustrating the danger of blocked fire exit routes in warehouses, accompanied by an annotated screenshot of the SARAS simulator.
  - *Cognitive State:* Interested; immediately recognises a challenge he faces daily in his own distribution centre.
  - *Action:* Clicks the link to view the SARAS Labs promotional website.
- **Frame 2: Landing Page Arrival (`/`):**
  - *Visual:* David lands on the clean, high-contrast homepage. The prominent hero headline reads: *"Workplace safety, learned by doing."* The supporting copy explains that SARAS provides accessible 360-degree hazard simulation without expensive VR headsets.
  - *Cognitive State:* Impressed by the professional presentation; relieved that no specialised hardware procurement is required.
  - *Action:* Clicks the primary CTA button: *"Explore Interactive Demo"*.
- **Frame 3: Educational Methodology & Architecture Review (`/how-it-works`):**
  - *Visual:* David explores the four-step pedagogical breakdown. He inspects the visual diagram contrasting passive induction videos with active spatial inspection and reviews the WCAG 2.2 AA accessibility specifications.
  - *Cognitive State:* Reassured that the platform is grounded in established instructional scaffolding and compliant with statutory HSE guidance.
  - *Action:* Proceeds directly to the interactive demo sandbox.
- **Frame 4: Direct Interactive Sandbox Engagement (`/demo`):**
  - *Visual:* Inside the embedded 16:9 viewport, David drags his mouse across a high-resolution 360-degree warehouse scene. He notices a pulsing reticle near an electrical distribution board obstructed by wooden pallets. He clicks the reticle, triggering a multiple-choice challenge. He deliberately selects an unsafe response to test the platform.
  - *System Response:* The platform immediately displays a high-contrast danger alert card (`#D62828`) with detailed explanatory text: *"Incorrect: Combustible pallets placed within 1 metre of an electrical panel create a severe flashover risk. Clear a minimum 1.0-metre safety perimeter immediately."*
  - *Cognitive State:* Delighted by the immediate explanatory feedback. Realises this will actively teach his new agency hires *why* rules exist.
- **Frame 5: Industrial Proof & Case Study Review (`/about`):**
  - *Visual:* David scrolls to the case study section detailing the industrial collaboration with Autoport Pvt. Ltd. and reads the verified quote from Mr. Ninson Lama confirming that the prototype helped streamline onboarding.
  - *Cognitive State:* Confident that the platform is tested in real industrial conditions and not merely an abstract academic exercise.
  - *Action:* Clicks *"Request 14-Day Free Pilot Access"*.

- **Frame 6: Pilot Booking & Conversion (`/contact`):**
  - *Visual:* David completes the simple, clean contact form, providing his work email, company name, and selecting "Logistics / 3PL" from the sector dropdown. He clicks *"Submit Pilot Request"*.
  - *System Response:* An instant green success notification confirms receipt. Within 4 hours, he receives a personalized email from Rijan Kapur Poudel offering two cohort trial links and scheduling a 15-minute Zoom orientation.
  - *Outcome:* Zero-cost inbound lead converted into an active industrial pilot deployment.

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13. Consistent Design Prototypes & Wireframe Analysis

To satisfy the highest criteria of the university marking brief, the team engineered high-fidelity design wireframes in Figma, ensuring visual consistency, modular component architecture, and responsive parity across all viewports.

13.1 Visual Design Prototype Integration

The submission portfolio incorporates the verified visual wireframe design assets created during the project design phase:



Figure 1: Full-length visual wireframe design prototype for the SARAS Labs promotional website, engineered in Figma by Sushank Jung Rawal.

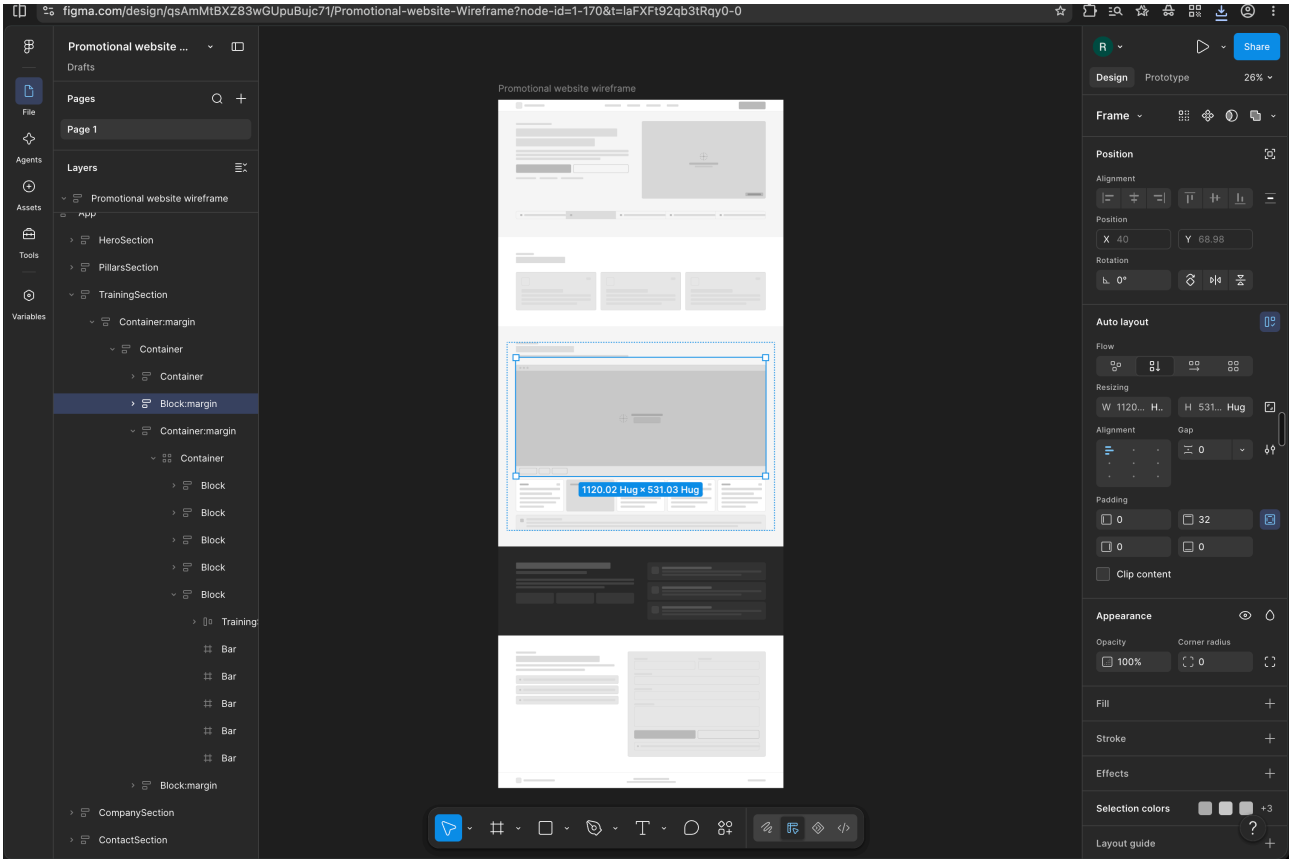


Figure 2: Verifiable documentary evidence of the promotional website wireframes, component design system, and responsive layout grids constructed directly within the Figma canvas.

13.2 Structural Wireframe Module Breakdown

The promotional website wireframe is architected into six distinct structural modules:

1. **Header Navigation Module:**
   - Dimensions: 1440 × 80 px (Desktop). Fixed elevation with `z-index: 100`.
   - Components: Vector logo mark, primary navigation links with active state indicator underlines (2px `#0056D2`), and a high-contrast primary CTA button with 12 px internal padding and 6 px corner radii.
2. **Hero Presentation Module:**
   - Dimensions: 1440 × 640 px. Two-column asymmetrical layout (55% copy, 45% visual).
   - Left Column: Primary brand tagline, value proposition copy, and dual CTA cluster.
   - Right Column: Browser viewport container framing the 360-degree warehouse scene with visual reticle callouts.
3. **Credibility & Standards Module:**
   - Dimensions: 1440 × 120 px. Light neutral background (`#F8F9FA`).
   - Components: 3-column horizontally aligned badges with vector icons representing HSE compliance, browser universality, and WCAG accessibility.
4. **Feature Triad & Curriculum Module:**
   - Dimensions: 1440 × 720 px. 12-column grid container.

- Components: Three elevated white cards (`#FFFFFF`) with 24 px padding, subtle drop shadows (`0 4px 12px rgba(11, 15, 20, 0.06)`), and 2px border radius. Displays Module 1 (Active), Module 2 (Roadmap), and Module 3 (Roadmap).

5. **Interactive Sandbox Preview Module:**
   - Dimensions: 1440 × 800 px. Dark neutral background (`#0B0F14`).
   - Components: Embedded 16:9 WebGL simulation frame with operator HUD, compass heading indicator, hotspot counters, and real-time explanatory modal overlay.
6. **Social Proof & Conversion Module:**
   - Dimensions: 1440 × 580 px. Two-column split layout.
   - Left Column: Autoport Pvt. Ltd. client testimonial card, quotation mark glyph, and industrial partner profile.
   - Right Column: Accessible 4-field pilot inquiry form with real-time validation states.

13.3 Responsive Breakpoint Specifications

The website wireframes adhere to a strict responsive grid system ensuring visual integrity across all modern computing devices:

Responsive Viewport Dimension	Grid Layout Structure	Margins & Gutters	Layout Adaptation & Component Behavior
Desktop Viewport (1440px+)	12-column fluid grid	80px outer margins 24px column gutters	Side-by-side hero layout; 3-column feature cards; persistent top navigation bar; large 16:9 embedded WebGL sandbox.
Tablet Viewport (768px – 1024px)	8-column fluid grid	40px outer margins 16px column gutters	Stacked hero section (headline above mockup); 2-column feature cards with third card spanning full width; compact header.
Mobile Viewport (375px – 480px)	4-column fluid grid	16px outer margins 12px column gutters	Single-column vertical stack; hamburger navigation drawer; sticky bottom CTA banner; touch-optimised 48 px button targets.

13.4 Touch Target & Interactive Accessibility Controls

- **Minimum Touch Sizing:** All interactive buttons, links, form inputs, and hotspot markers maintain a strict minimum dimension of 48 × 48 px in accordance with WCAG 2.2 Success Criterion 2.5.8 (Target Size).
- **Focus State Visibility:** All focusable elements display a clear, 2px solid SARAS Primary Blue (`#0056D2`) focus ring with a 2px offset, ensuring effortless keyboard navigation for evaluators using the `Tab` key.
- **Semantic HTML Markup:** Form elements feature explicitly linked `<label>` tags, ARIA live regions for validation errors, and descriptive `alt` attributes on all diagrammatic assets.

References & Academic Bibliography

- Nielsen, J. (1994, updated 2024) *10 Usability Heuristics for User Interface Design*. Nielsen Norman Group. Available at: https://www.nngroup.com/articles/ten-usability-heuristics/.
- University of Sunderland (2024) *CET257 Enterprise Project Assignment 1 Brief & Marking Criteria*. Sunderland: School of Computer Science.
- World Wide Web Consortium (W3C) (2024) *Web Content Accessibility Guidelines (WCAG) 2.2*. W3C Recommendation. Available at: https://www.w3.org/TR/WCAG22/.